

Chapter 5

Computer Vision

Introduction

Agriculture has played an important role in the global economy in recent years. The further growth of the population is leading to a gradual decrease in the area of the agricultural land and further growing stresses on the farm system. Agricultural food production methods are rising in demand for successful and healthy. Innovative sensing and motive technologies and advanced information and communications systems and Artificial Intelligence must supplement conventional agricultural management methods to speed up farm productivity growth more accurately, thereby promoting the production of high-quality and high-performance agriculture. Computer vision monitoring systems have become valuable tools in farm operations over the past decades, and their use has increased significantly.

What is Computer Vision?

Computer vision can be characterized as a field of AI for extracting information from digital images. The type of data obtained from an image may be different from recognition, navigational space measurements or applications of increased reality.

The applications of computer vision can also be described. Computer vision builds algorithms that recognize and use the quality of pictures for other purposes.

People look at and physically experience the world around them through the eyes and the brains. Computer vision is the science that wants to provide a machine or a device with similar, if not better, performance.

The aim of computer vision is to automatically collect, interpret and understand useful and meaningful data from a single image or a series of images. This consists of establishing a theoretical and algorithmic basis for automatic visual comprehension.

In agriculture, Computer vision systems are clearly feasible for grading and evaluating properties such as color, form, thickness, surface defects and contamination of food products into different grades.

Computer Vision hierarchy: Computer vision is classified into three basic categories:

- Low Level : It contains a process image for extraction of features
- Intermediate Level: This involves recognition of objects and 3D scenes
- High level: It provides a conceptual overview of an incident, goal and comportsment scenario

Task of Computer Vision

- **Recognition:** The normal issue in computer vision, image processing and machine vision is whether or not the image data contains any specific object, function or operation. Normally, a person can overcome this dilemma effectively and easily, however in the general case machine view it is still not satisfactorily solved: arbitrary objects in arbitrary circumstances. The available methods of dealing in this field can at best solve this problem only for particular objects, such as basic geometric objects (e.g., polyhedrons), human features, written or manuscript characters or vehicles, and surroundings in unique circumstances that are usually well described in terms of illumination.

There are different variety of Recognition come to notice like

Recognition: One or more objects or classes, usually together with their 2D image location or 3D poses, can be recognized pre-specified or learned in a given scene. It can be any person, item like fruits, vegetables etc.

Identification: A separate instance is recognized for an object. *Examples:* identification of the face or fingerprint of a specific individual or identification of a particular vehicle.

Detection: For a specific situation the image data is screened. *Examples:* identification, in agricultural images, of potential irregular cells or tissues or automatic road toll identification. Detection by comparatively simple and quick calculation is often used to identify small regions with interesting data, which can be analyzed further in order to obtain a correct interpretation by using more computationally challenging technologies.

- **Motion Analysis:** Various tasks involve the movement estimation when processing an image sequence to measure the speed at each point either in the image or the 3D image scene, or even the camera that produces the images.

Examples are the following:

Egomotion: Determination of the 3D rigid motion of the camera from the camera image series (rotation and translation).

Tracking: Following (normally) movement of a series of points of interest or objects in the picture sequence (e.g., cars, human beings or other organisms).

Optical flow: to evaluate how the point moves in relation to the image plane for each point in the image, i.e., its apparent motion. This move represents both the way the 3D point moves in the scene and the way the camera moves in relation to the object.

- **Scene Reconstruction:** If a scene reconstruction has one or (typically) more images of a scene or a picture, it is intended to compute a 3D model of the scene. The model may be a series of 3D points in the simplest case. More detailed methods produce a complete 3D surface model. The introduction of 3D imagery without motion or scanning and the related processing algorithms allows fast progress in this area. 3D grid sensing can be used in multi-angle processing of three-dimensional images. Several 3D images are now available to be merged into point clouds and 3D models

- **Image Restoration:** The goal of image restore is to eliminate noise from photographs (sensor noise, motion blur, etc). Diverse filter types such as low-pass filters or medium filters are the easiest possible solution for noise reduction. More advanced approaches are designed to differentiate between noise and local image structures. When the image data are first analyzed by local image structures, such as lines or edges, and then the filtering is managed based on local information from the analysis stage, a better noise removal level is usually achieved compared with simpler approaches. The recognition/detection of images is correlated with many challenges:

- viewpoint variation
- scale variation
- intra-class variation
- image deformation
- image occlusion
- illumination
- background clutter

Computer Vision System: Computer vision system structure is highly important. In some cases the device is a subsystem of a larger design, for example with subsystems for mechanical actuator control, planning, repositories of information, man-machine interfaces, etc. Some systems have separate implementations, which solve a particular measuring and detection issue. This would also depend on whether its functionality is pre-specified or whether part of the computer vision program can be updated or amended during service. However, many computer vision systems have traditional features.

Image Acquisition: Capturing of One or more images using digital image sensors which are developed to include range sensors, tomography equipment, radar, ultrasonic cameras, etc., in addition to specific type of light sensitive cameras. The resulting image data depends on the form of sensor: a specific 2D image, a 3D volume or a series of images. The pixel values usually refer to the intensity of light in one or more spectral lengths (gray images or color images), but can also apply to various physical measurements, such as sonic or electromagnetic wave width, absorption or reflectance, or the nuclear magnet resonance.

Pre-Processing: In order to retrieve certain pieces of information, a Computer vision approach may usually be used to analyze the data to ensure that all expectations suggested by the technique can be fulfilled.

Feature Extraction: The image data extracts feature at various levels of complexity. Typical examples are the following: lines, edges and ridges. Localized points of interest like edges, blobs or lines. The texture, form or movement can be more complex.

Detection/Segmentation: There is a decision at some stage in the course of the processing which image points or image areas are important for further care. which can be like:

- Order of a particular set of points of interest.
- Segmentation of one or more parts of the image containing a specific point of interest.
- Splitting the image into the nested scene Architecture consisting of the first-ground, object classes, individual objects or salient object pieces (also referred to as the hierarchy of the spatial taxon scene).
- segmenting or co-segmenting one or more videos to a set of foreground masks per frame while preserving its time semantical continuity

High Level Processing: The input typically consists of a small data set, such as a set of points or a picture area which is supposed to contain one particular object. It deals with

- Verification that the data meet clear assumptions based on model and implementation.
- Application-specific parameters such as object position or object size are estimated.
- Image recognition – grouping into various categories of a detected object.
- Image registration – comparing and integrating the same object’s two different views.
- **Decision Making:** Giving the final judgment for the submission,
 - *Example*, Automated testing applications pass/fail.
 - Match/no-match in applications for recognition.
 - Additional human review flag in applications for medical, military, protection and recognition.

For the Implementation of Computer vision, we use framework like OpenCV. Which we will study in next section.

OPENCV

OpenCV is an opensource package or library which is developed by Intel. This library is cross platform it means we can use it with java, C++, python etc., In all kind of image and video processing and analysis, the OpenCV or the Open Source Control Vision software (<http://opencv.org>) is used. It can process images and videos to recognize people, faces, or even manuscripts. Integrated into many libraries, including NumPy, Python is able to analyze the OpenCV array structure. OpenCV converts visual knowledge machine vision to the space of the object. The image pattern and functionality from the vector space can be defined and mathematical operations are carried out on these features.

Installing OpenCV

On Windows: Below is a list of the steps needed to complete the installation.

Install OpenCV: Run the command “pip install opencv_python” On your Command prompt of windows and OpenCV will be automatically get installed.

On linux/ubuntu:

Importing OpenCV: Get to the IDLE and import OpenCV:

```
>>> import cv2  
You can also check which version you have:  
>>> cv2.__version__  
'3.4.1'
```

Note: Version can be different it depends on its release

Let's try something with OpenCV:

Example save it as opencv.py:

```
import cv2 #OPENCV library  
# Load an color image  
img = cv2.imread ('download.jpg')  
#Image should be in the same folder where program file is saved  
cv2.imshow ('image',img)  
cv2.waitKey (0)  
cv2.destroyAllWindows ()
```

Output:

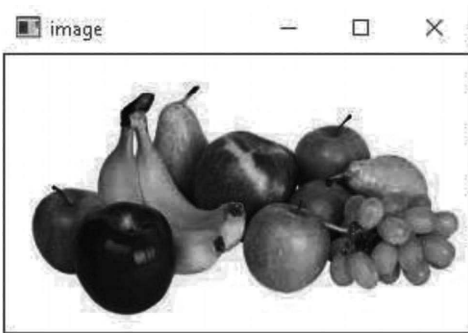


Fig. 5.1. Output Image

We can convert the above Fig 5.1 image into grey scale by passing value 0 into the argument img

```
import cv2 #OPENCV library  
# Load an color image  
img = cv2.imread ('download.jpg',0)  
# passing 0 to convert it into grey scale  
#Image should be in the same folder where program file is saved  
cv2.imshow ('image',img)  
cv2.waitKey (0)  
cv2.destroyAllWindows ()
```

Output:

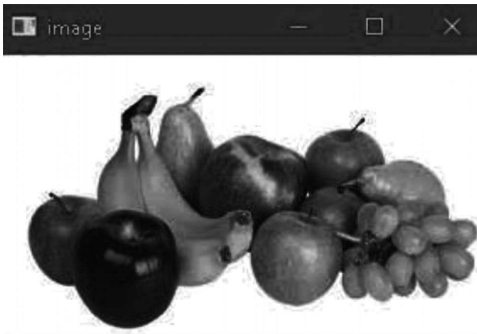


Fig. 5.2 Grey Image

Explanation: `Cv2.imread ()` is used to read an image. And as seen by passing second argument we have converted image to grey scale as seen in Fig 5.2.

The second arguments always depict how the image should be read:

- **`cv2.IMREAD_COLOR`:** Loads an image color. Any image transparency is overlooked. It's the predetermined mark.
- **`cv2.IMREAD_GRAYSCALE`:** Loads image in grayscale
- **`cv2.IMREAD_UNCHANGED`:** Loads image as such including alpha channel.
- **`cv2.imshow ()`:** used to display an image. The output window automatically fit to the size of an image. The first argument is a string window name. Our photo is the second statement. You can build as many windows, but with different names of windows as you want.
- **`cv2.waitKey ()`:** The keyboard binding feature in OpenCV. The time in milliseconds is the point. For each keyboard case, this feature is waiting for milliseconds. The software continues if you click a key during that time. If 0 is reached, it waits for a central stroke forever. You may also set it to detect other keystrokes such as, when key an is pressed, etc.
- **`cv2.destroyAllWindows ()`:** It destroys all the windows created and if any particular window is to be destroyed, we can use the function `cv2.destroywindow ()` and pass the window name as an argument into it.

OpenCV is widely used in so many domains as an important part of various technologies and we will use open cv in our further chapters for various purpose and a lot of other topics will be discussed. Now let's make an real time project with the help of OpenCV.

Crop Grading

In agriculture crop quality grading is a very important thing as it provides the simple by using OpenCV

```
import cv2
import numpy as np
from matplotlib import pyplot as plt

def get_classification (ratio):
    ratio =round (ratio,1)
    toret=""
    if (ratio>=3):
        toret="Slender"
    elif (ratio>=2.1 and ratio<3):
        toret="Medium"
    elif (ratio>=1.1 and ratio<2.1):
        toret="Bold"
    elif (ratio<=1):
        toret="Round"
    toret=" (" +toret+)"
    return toret

#rnjn
print ("Starting")
img = cv2.imread ('rice.png',0) #Path where images is stored
#load in greyscale mode

#convert into binary
ret,binary = cv2.threshold (img,160,255,cv2.THRESH_BINARY)# 160 -
threshold, 255 - value to assign, THRESH_BINARY_INV - Inverse
binary

#averaging filter
kernel = np.ones ( (5,5),np.float32)/9
dst = cv2.filter2D (binary,-1,kernel)# -1 : depth of the destination
image

kernel2 = cv2.getStructuringElement (cv2.MORPH_ELLIPSE, (3,3))

#erosion
erosion = cv2.erode (dst,kernel2,iterations = 1)

#dilation
dilation = cv2.dilate (erosion,kernel2,iterations = 1)

#edge detection
```

```

edges = cv2.Canny (dilation,100,200)

### Size detection
_,contours,hierarchy = cv2.findContours (erosion, cv2.RETR_
EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
print ("No. of rice grains=",len (contours))
total_ar=0
for cnt in contours:
    x,y,w,h = cv2.boundingRect (cnt)
    aspect_ratio = float (w)/h
    if (aspect_ratio<1):
        aspect_ratio=1/aspect_ratio
    #print (round (aspect_ratio,2),get_classificaton
        (aspect_ratio))

    total_ar+=aspect_ratio
avg_ar=total_ar/len (contours)
print ("Average Aspect Ratio=",round (avg_ar,2),get_classificaton
        (avg_ar))

#plot the images
imgs_row=2
imgs_col=3
plt.subplot (imgs_row,imgs_col,1),plt.imshow (img,'gray')
plt.title ("Original image")

plt.subplot (imgs_row,imgs_col,2),plt.imshow (binary,'gray')
plt.title ("Binary image")

plt.subplot (imgs_row,imgs_col,3),plt.imshow (dst,'gray')
plt.title ("Filtered image")

plt.subplot (imgs_row,imgs_col,4),plt.imshow (erosion,'gray')
plt.title ("Eroded image")

plt.subplot (imgs_row,imgs_col,5),plt.imshow (dilation,'gray')
plt.title ("Dialated image")

plt.subplot (imgs_row,imgs_col,6),plt.imshow (edges,'gray')
plt.title ("Edge detect")

```



```
plt.show ()
```

Output:

```
>>>
```

```
RESTART:      E:\studies\Projects\BOOK\Rice\rice-quality-analysis-  
master\code.py
```

```
Starting
```

```
No. of rice grains= 30
```

```
Average Aspect Ratio= 2.3 (Medium)
```

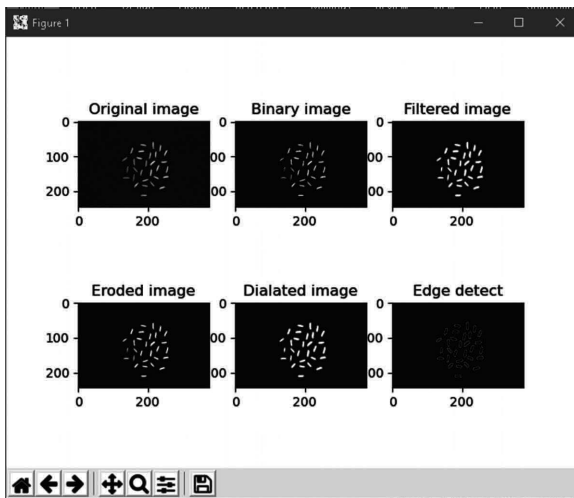


Fig. 5.3 Output of Rice using Matplotlib

As we can see in the Fig 5.3 how rice has been detected and grading has been done using OpenCV which is represented using matplotlib library which is used for visualization of output after grading of rice.

Summary

Computer Vision is considerably the eyes of artificial Intelligence which can help to process any video or pictures and understand them for generating a rightful data. In this chapter we have seen the importance of computer vision and how it can be useful in the agriculture for identification and with two small example we have also demonstrated the use of computer vision.